# Large values of the Gowers-Host-Kra seminorms

Terence Tao · 16 December 2010

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Tanja Eisner and I have just uploaded to the arXiv our paper "Large values of the Gowers-Host-Kra seminorms", submitted to Journal d'Analyse Mathematique. This paper is concerned with the properties of three closely related families of (semi)norms, indexed by a positive integer $k$:

- The *Gowers uniformity norms* $\|f\|_{U^k(G)}$ of a (bounded, measurable, compactly supported) function $f: G \to \mathbf{C}$ taking values on a locally compact abelian group $G$, equipped with a Haar measure $\mu$;
- The Gowers uniformity norms $\|f\|_{U^k([N])}$ of a function $f: [N] \to \mathbf{C}$ on a discrete interval $\{1, \ldots, N\}$; and
- The Gowers-Host-Kra seminorms $\|f\|_{U^k(X)}$ of a function $f \in L^\infty(X)$ on an ergodic measure-preserving system $X = (X, \mathcal{X}, \mu, T)$.

These norms have been discussed in depth in previous blog posts, so I will just quickly review the definition of the first norm here (the other two (semi)norms are defined similarly). The $U^k(G)$ norm is defined recursively by setting

$$\|f\|_{U^1(G)} := \left| \int_G f \, d\mu \right|$$

and

$$\|f\|_{U^k(G)}^{2^k} := \int_G \|\Delta_h f\|_{U^{k-1}(G)}^{2^{k-1}} \, d\mu(h)$$

where $\Delta_h f(x) := f(x+h)\overline{f(x)}$. Equivalently, one has

$$\|f\|_{U^k(G)} := \left( \int_G \ldots \int_G \Delta_{h_1} \ldots \Delta_{h_k} f(x) \, d\mu(x) d\mu(h_1) \ldots d\mu(h_k) \right)^{1/2^k}.$$

Informally, the Gowers uniformity norm $\|f\|_{U^k(G)}$ measures the extent to which (the phase of $f$) behaves like a polynomial of degree less than $k$. Indeed, if $\|f\|_{L^\infty(G)} \leq 1$ and $G$ is compact with normalised Haar measure $\mu(G) = 1$, it is not difficult to show that $\|f\|_{U^k(G)}$ is at most 1, with equality if and only if $f$ takes the form $f = e(P) := e^{2\pi i P}$ almost everywhere, where $P: G \to \mathbf{R}/\mathbf{Z}$ is a polynomial of degree less than $k$ (which means that $\partial_{h_1} \ldots \partial_{h_k} P(x) = 0$ for all $x, h_1, \ldots, h_k \in G$).

Our first result is to show that this result is robust, uniformly over all choices of group $G$:

**Theorem 1 ($L^\infty$-near extremisers)** Let $G$ be a compact abelian group with normalised Haar measure $\mu(G) = 1$, and let $f \in L^\infty(G)$ be such that $\|f\|_{L^\infty(G)} \leq 1$ and $\|f\|_{U^k(G)} \geq 1 - \epsilon$ for some $\epsilon > 0$ and $k \geq 1$. Then there exists a polynomial $P : G \to \mathbf{R}/\mathbf{Z}$ of degree at most $k-1$ such that $\|f - e(P)\|_{L^1(G)} = o(1)$, where $o(1)$ is bounded by a quantity $c_k(\epsilon)$ that goes to zero as $\epsilon \to 0$ for fixed $k$.

The quantity $o(1)$ can be described effectively (it is of polynomial size in $\epsilon$), but we did not seek to optimise it here. This result was already known in the case of vector spaces $G = \mathbf{F}_p^n$ over a fixed finite field $\mathbf{F}_p$ (where it is essentially equivalent to the assertion that the property of being a polynomial of degree at most $k-1$ is locally testable); the extension to general groups $G$ turns out to fairly routine. The basic idea is to use the recursive structure of the Gowers norms, which tells us in particular that if $\|f\|_{U^k(G)}$ is close to one, then $\|\Delta_h f\|_{U^{k-1}(G)}$ is close to one for most $h$, which by induction implies that $\Delta_h f$ is close to $e(Q_h)$ for some polynomials $Q_h$ of degree at most $k-2$ and for most $h$. (Actually, it is not difficult to use cocycle equations such as $\Delta_{h+k} f = \Delta_h f \times T^h \Delta_k f$ (when $|f| = 1$) to upgrade "for most $h$" to "for all $h$".) To finish the job, one would like to express the $Q_h$ as derivatives $Q_h = \partial_h P$ of a polynomial $P$ of degree at most $k-1$. This turns out to be equivalent to requiring that the $Q_h$ obey the cocycle equation

$$Q_{h+k} = Q_h + T^h Q_k$$

where $T^h F(x) := F(x+h)$ is the translate of $F$ by $h$. (In the paper, the sign conventions are reversed, so that $T^h F(x) := F(x-h)$, in order to be compatible with ergodic theory notation, but this makes no substantial difference to the arguments or results.) However, one does not quite get this right away; instead, by using some separation properties of polynomials, one can show the weaker statement that

$$Q_{h+k} = Q_h + T^h Q_k + c_{h,k} \tag{1}$$

where the $c_{h,k}$ are small real constants. To eliminate these constants, one exploits the trivial cohomology of the real line. From (1) one soon concludes that the $c_{h,k}$ obey the 2-cocycle equation

$$c_{h,k} + c_{h+k,l} = c_{h,k+l} + c_{k,l}$$

and an averaging argument then shows that $c_{h,k}$ is a 2-coboundary in the sense that

$$c_{h,k} = b_{h+k} - b_h - b_k$$

for some small scalar $b_h$ depending on $h$. Subtracting $b_h$ from $Q_h$ then gives the claim.

Similar results and arguments also hold for the $U^k([N])$ and $U^k(X)$ norms, which we will not detail here.

Dimensional analysis reveals that the $L^\infty$ norm is not actually the most natural norm with which to compare the $U^k$ norms against. An application of Young's convolution inequality in fact reveals that one has the inequality

$$\|f\|_{U^k(G)} \leq \|f\|_{L^{p_k}(G)} \tag{2}$$

where $p_k$ is the critical exponent $p_k := 2^k/(k+1)$, without any compactness or normalisation hypothesis on the group $G$ and the Haar measure $\mu$. This allows us to extend the $U^k(G)$ norm to all of $L^{p_k}(G)$. There is then a stronger inverse theorem available:

> **Theorem 2 ($L^{p_k}$-near extremisers)** Let $G$ be a locally compact abelian group, and let $f \in L^{p_k}(G)$ be such that $\|f\|_{L^{p_k}(G)} \leq 1$ and $\|f\|_{U^k(G)} \geq 1-\epsilon$ for some $\epsilon > 0$ and $k \geq 1$. Then there exists a coset $H$ of a compact open subgroup $H$ of $G$, and a polynomial $P : Hto\mathbf{R}/\mathbf{Z}$ of degree at most $k-1$ such that $\|f - e(P)1_H\|_{L^{p_k}(G)} = o(1)$.

Conversely, it is not difficult to show that equality in (2) is attained when $f$ takes the form $e(P)1_H$ as above. The main idea of proof is to use an inverse theorem for Young's inequality due to Fournier to reduce matters to the $L^\infty$ case that was already established. An analogous result is also obtained for the $U^k(X)$ norm on an ergodic system; but for technical reasons, the methods do not seem to apply easily to the $U^k([N])$ norm. (This norm is essentially equivalent to the $U^k(\mathbf{Z}/\tilde{N}\mathbf{Z})$ norm up to constants, with $\tilde{N}$ comparable to $N$, but when working with near-extremisers, norms that are only equivalent up to constants can have quite different near-extremal behaviour.)

In the case when $G$ is a Euclidean group $\mathbf{R}^d$, it is possible to use the sharp Young inequality of Beckner and of Brascamp-Lieb to improve (2) somewhat. For instance, when $k=3$, one has

$$\|f\|_{U^3(\mathbf{R}^d)} \leq 2^{-d/8}\|f\|_{L^2(\mathbf{R}^d)}$$

with equality attained if and only if $f$ is a gaussian modulated by a quadratic polynomial phase. This additional gain of $2^{-d/8}$ allows one to pinpoint the threshold $1-\epsilon$ for the previous near-extremiser results in the case of $U^3$ norms. For instance, by using the Host-Kra machinery of characteristic factors for the $U^3(X)$ norm, combined with an explicit and concrete analysis of the 2-step nilsystems generated by that machinery, we can show that

$$\|f\|_{U^3(X)} \leq 2^{-1/8}\|f\|_{L^2(X)}$$

whenever $X$ is a *totally* ergodic system and $f$ is orthogonal to all linear and quadratic eigenfunctions (which would otherwise form immediate counterexamples to the above inequality), with the factor $2^{-1/8}$ being best possible. We can also establish analogous results for the $U^3([N])$ and $U^3(\mathbf{Z}/N\mathbf{Z})$ norms (using the inverse $U^3$ theorem of Ben Green and myself, in place of the Host-Kra machinery), although it is not clear to us whether the $2^{-1/8}$ threshold remains best possible in this case.

## Comments (3)



**Anon** · 5 October 2018

Dear Terence Tao, could you please indicate some more detail on how one deduces from (1) the 2-cocyle equation c_{h,k} + c_{h+k,l} = c_{h,k+l} + c_{k,l} ? In your paper with Eisner I cannot see any details either for how to do this deduction.



**Terence Tao** · 5 October 2018

From (1) we have $Q_{h+k+l} = Q_h + T^h Q_{k+l} + c_{h,k+l}$, $Q_{k+l} = Q_k + T^k Q_l + c_{k,l}$, $Q_{h+k+l} = Q_{h+k} + T^{h+k} Q_l + c_{h+k,l}$, and $Q_{h+k} = Q_h + T^h Q_k + c_{h,k}$. Use these identities to expand out $Q_{h+k+l}$ in two different ways and collect terms.



**Anonymous** · 6 October 2018

Great, thank you!